

What are the sample_weight parameter in various performance metrics

Asked 4 years, 10 months ago Modified 4 years, 10 months ago Viewed 2k times

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Can someone explain what is the sample_weight parameter for in various scikit-learn performance metrics such as:

- accuracy_score (https://scikit-learn.org/stable/modules/generated/sklearn.metrics.accuracy_score.html)
- Jaccard_score (https://scikit-learn.org/stable/modules/generated/sklearn.metrics.jaccard_score.html#sklearn.metrics.jaccard_score)
- F1-score (https://scikit-learn.org/stable/modules/generated/sklearn.metrics.f1_score.html) etc.

Many thanks.

python

performance

scikit-learn

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asked Jan 25, 2020 at 9:27
 [Leockl](#)
2,156 6 26 56

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It simply indicates how much each sample affects the metric. Look at this example:

With below y_true and y_pred the accuracy_score would be 0.6666666 :

```
y_true = [0, 0, 1]  
y_pred = [0, 1, 1]  
  
accuracy_score(y_true, y_pred) # 0.6666666666666666
```

But if second sample is more important to us than other two, we can enforce it's significance with sample_weight :

```
accuracy_score(y_true, y_pred, sample_weight=[1, 2, 1]) # 0.5
```

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answered Jan 25, 2020 at 10:46
 [Mehraban](#)
3,304 5 39 63

- Many thanks for explaining this and including an example! This makes sense. Do you know in what situations/scenarios we would want to adjust the sample_weights?
– [Leockl](#) Jan 25, 2020 at 10:59
- 1

most of all in case of imbalanced data in which you have significantly less samples in some classes, it would be good to know if model has bias toward classes with more samples or not. – [Mehraban](#) Jan 25, 2020 at 11:02